Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: **Kataribe**
b. Team leader name: **Tanguy Pemeja**
c. Problem Statement: **Sustainability & Social Impact**

Brief about the idea

Kataribe 語り部

Every grandparent has stories nobody has asked about yet.

Kataribe talks with an elderly person for a few minutes a day, in Japanese, and turns what they remember into a life story their family can read, and hear in their own voice.

THE PROBLEM

Many Japanese seniors spend their days alone, and their families never find the time or the courage to ask about their lives. The stories go unrecorded.

WHO IT'S FOR

Them: a few minutes a day about their own life. Their family: the stories they never asked for. Next, their town: a friend, a carer, a local archive.

NOT A COMPANION

It does not stand in for anyone. A kataribe is a keeper of oral history; this one helps the people around them know them better.

6.72M

people aged 65 and over live alone in Japan

2020 census · Cabinet Office, Annual Report on the Ageing Society

¥220,000

for one professionally written self-history (自分史), made once, by hand

Oya no Zasshi (親の雑誌, Kokoromi), price incl. tax

Opportunities

Not a companion that stands in for anyone: a way for the people around them to know them better.

How different is it from any of the other existing ideas?

- Memoir AIs (Tell Mel, Storyworth, Remento) rewrite a parent's words into chapters, and the family chooses who receives them.
- Companion AIs (Cotomo, by Starley) keep seniors company, but no life story comes out.
- A human-written self-history costs ¥220,000, and is made once.

How will it be able to solve the problem?

- A few minutes a day in natural Japanese. The interviewer speaks first: nothing to install or learn.
- Each conversation becomes stories, set as a book by time of life. What is untold plans the next.
- Relatives read what was shared, hear the voice behind each quote, and send questions it asks next.

USP of the proposed solution

- Theirs to give: nothing reaches the family without a yes in their own words, and any story can be taken back.
- It retells, but never adds: every fact has a quote, found word for word, then heard in the recording.
- Built for Japanese: honorific register, names spelled from the profile, patience for a pause.

List of features offered by the solution

Talks in Japanese
Gemini Live, voice to voice. Speaks first, asks one thing, waits 5 s through a pause.

Remembers last time
Opens by recalling the last conversation, or with a question the family sent.

A life story, as a book
Chapters by time of life. Each story retold, its quotes a press away, each with its clip.

Never adds a detail
Every fact retold has a quote; each quote found word for word, then heard in its clip.

Plans from the gaps
Finds the times of life not yet told, and plans the next three questions.

Asked at the close
Shared only on a yes in their own words. No reply: asked again. Any story can be taken back.

The family asks back
A relative says who they are and asks; it says who asked. Refused topics are never raised.

Japanese, with English
The family page in one language at a time, with a switch. Quotes keep the words as said.

Process flow diagram or Use-case diagram

Every conversation

The family

1 Talk
Live, by voice. Speaks first, asks one thing, waits through a pause.

2 Write
The talk becomes stories, each resting on their own sentences.

3 Check
Each quote found word for word, then heard in the recording.

4 Share
At the close: may the family read today's? Only a yes goes on.

5 Plan
Finds what is still untold; plans the next three questions.

Next conversation
Opens with last time, or with the family's question.

next time

only what was shared

Family reads
A book by time of life; each quote with its clip.

Family asks back
A relative says who they are, then asks: grandchild Yuka.

Screened
A question touching a refused topic is quietly held back.

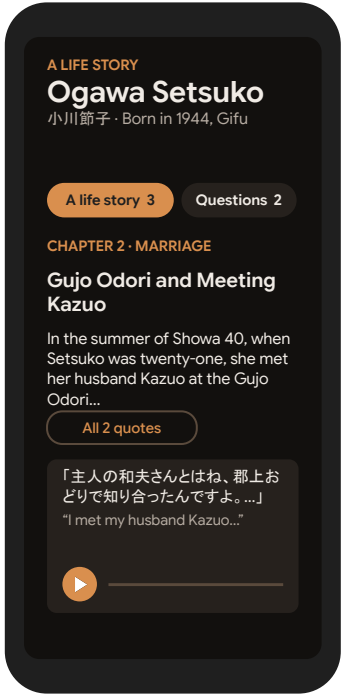
two at most

Wireframes/Mock diagrams of the proposed solution (optional)



The senior's screen: one button to start, nothing to learn

1. Its words, large, as it says them
2. Listening or speaking, calmly
3. One button to finish
4. Then: share today's story, or keep it

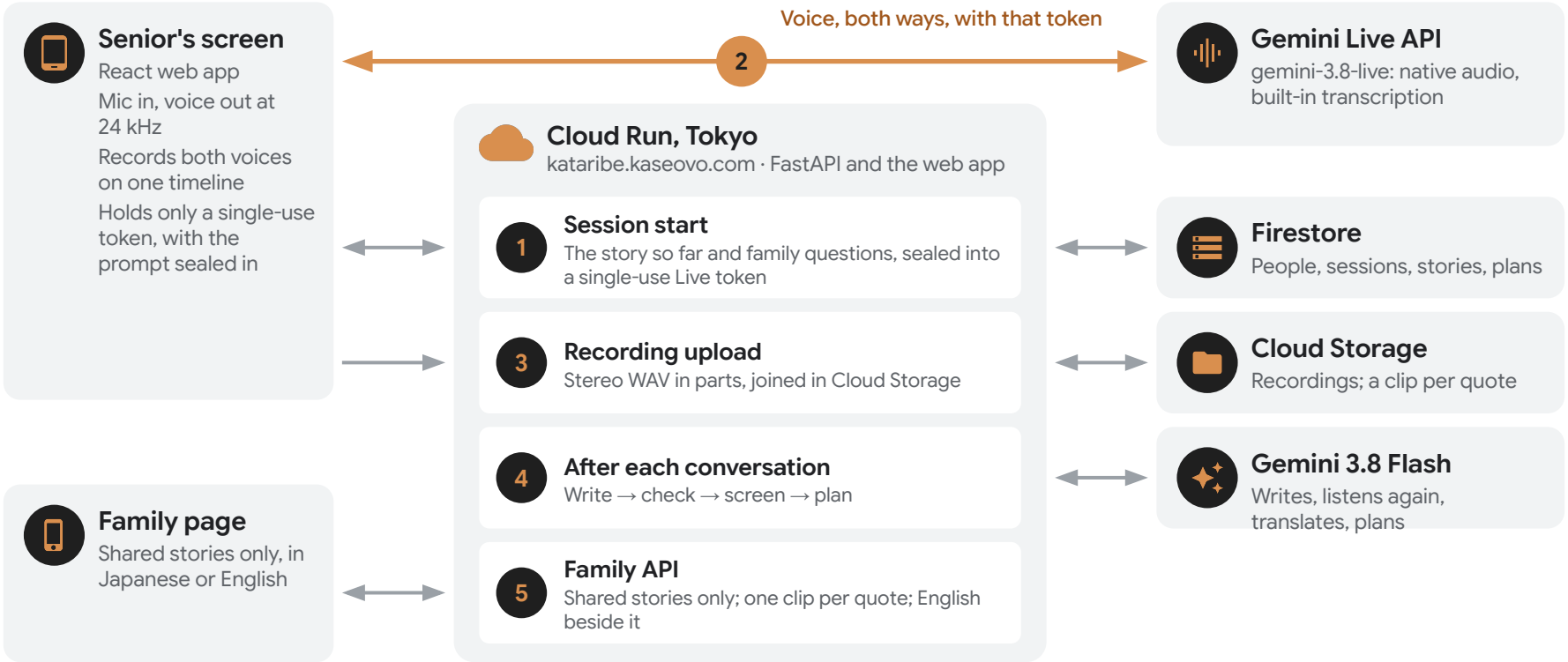


The family page

1. The life story as a book, by time of life
2. The family's questions, one tab away, each with the story it led to
3. Each retelling with its quotes, in the words as said, and the voice that said them

Both are built. kataribe.kaseovo.com shows a family page from the demo, and lets anyone try the conversation in English.

Architecture diagram of the proposed solution



Technologies to be used in the solution

Google AI

- Gemini 3.8 Live: real-time voice, native audio, built-in transcription
- Gemini 3.8 Flash: writes the stories, listens to each quote's clip, translates, plans
- Google Gen AI SDK, Python and JavaScript; single-use tokens

Google Cloud

- Cloud Run in Tokyo, at kataribe.kaseovo.com
- Firestore and Cloud Storage; Secret Manager for the keys
- An uptime check, Error Reporting and a budget alert

Application

- React 19, Vite, TypeScript
- Web Audio worklet: capture, and both voices on one recording
- FastAPI on Python 3.13, managed with uv

Quality

- pytest and Playwright against the live API, with synthesized Japanese speech
- AI-vs-AI sparring: three personas, real audio both ways
- make smoke: a judge's whole path, on the live service

One rule across the stack The model never speaks for the senior. Gemini writes the stories; plain code checks every quote and every yes against the transcript, and Gemini hears each one again in the recording.

Estimated implementation cost (optional)

≈ $0.13

per 10-minute conversation, in AI and cloud, at list prices

≈ ¥6,900

for a year of daily conversations, per person

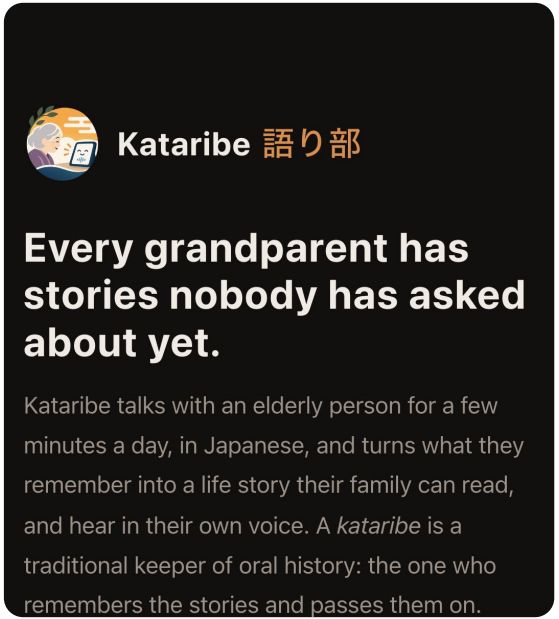
About 3% of one ¥220,000 self-history written by hand.

Item	Basis	Per session
Gemini Live, audio in	$0.005 a minute, for all 10 minutes	$0.050
Gemini Live, audio out	$0.018 a minute, about 2.5 minutes spoken	$0.045
Gemini Flash, writing	≈ 15k tokens in, 3k out: write, screen, plan	$0.023
Gemini Flash, listening	≈ 3 minutes of quoted clips, heard again	≈ $0.006
Recording storage	58 MB of WAV, at ≈ $0.023 per GB-month	< $0.002
Cloud Run, Firestore	Scale to zero; free tiers at this scale	≈ $0
Total		**≈ $0.13**

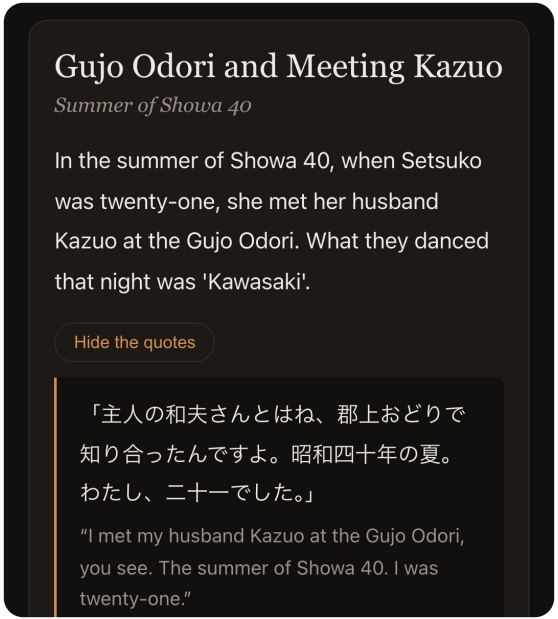
Audio is billed for the whole time a session is open, silence included. Rates are Google's list prices as of 23 Sep 2026, kept in backend/costs.py; token counts are estimates, the listening at 32 audio tokens a second. Yen at ¥150 to the dollar. Judging: at most 30 visitor conversations a day, about $0.10 each.

Snapshots of the prototype

Live at kataribe.kaseovo.com, on a phone: the demo's family page, written from the video's conversations.



The start page
What it is, who it's for, a family page from the demo, then a judge's own turn.



A story, and what it rests on
A retelling to read; every quote a press away, in Japanese with the English beneath.



Questions from the family
Who asked, and the story each question led to. A waiting one says so.

Prototype Performance report/Benchmarking

36 / 36
Consent, twelve cases: a yes shares; a no or a hesitation keeps it private; no reply is asked again

12 / 12
The recording check: the same words confirmed, 18 for 十八 accepted, a changed place caught

3 / 3
A judge's whole path on the live service, from the start page to their own family page

6 min
A browser session with no drop, 34 transcript entries

How long it waits before replying

Patience setting	Pause in speech	Result
3,500 ms	1.5 s	Waited
3,500 ms	2.5 s	Cut in
6,000 ms	2.5 s	Waited
6,000 ms	4.0 s	Waited

Japanese: 5,000 ms, measured. English visitors: 4 s. The model's own: 800 ms.

How it is measured

- Real models, three runs a case: speech streamed at the Live API in real time, replies graded by rules and by a model judging them against the prompt.
- AI-vs-AI sparring: three personas press a long pause, a refused topic, a change of subject.
- The recording check caught a real slip on its own: 私は transcribed as 私を. The quote was held back.
- Not yet measured: elderly voices and dialect. A native speaker's review of the Japanese is next.

Additional Details/Future Development (if any)

Next, on the same screen

- It notices the senior sitting nearby, so there is no button at all
- Names it is unsure of kept in kana; the family fixes the spelling in one tap
- A message to the family when a new story lands

New channels, the same brain

- Phone and landline
- Smart speakers
- Inside the companion robots Japanese families already buy

Beyond the family

- The teller chooses who else may read: a friend, a care worker, a local archive
- Testimony, the oldest sense of 語り部: the war, the bombings, the disasters, told by those who lived them
- Municipalities as payers: Yokosuka City has trialled conversation AI for seniors since 2025

How it scales Every channel plugs into the same backend. Cloud Run scales to zero and out. Gemini moves to Vertex AI in a Japanese region, behind one adapter, when data residency matters.

Provide links to your:

- GitHub Public Repository: github.com/kaseovo/kataribe
- Demo Video Link (3 Minutes): youtu.be/wg3SdraT2Oo
- Final Product Link: kataribe.kaseovo.com



Watch the demo

3:48, in English: two conversations, the family page they made, and how to try it yourself.



Try it on your own life

A family page from the demo, then a conversation of your own, in English. About three minutes.



The code, the tests, and the plan behind them

Open source, under the MIT licence. Built in the open from the first commit on 22 Sep 2026.

Community benefit and business case

For them

- A few minutes a day that are theirs, about their own life
- Remembering out loud, with someone who waits for a name or a year
- Theirs to give: nothing reaches the family until they say so

For their family

- The stories they never asked for: not for lack of love, for lack of a moment
- In their own words, with their voice behind each quote
- Something to talk about, and a question to send

For their town, next

- Whoever they choose: a friend, a care worker, a local archive
- Many of the 6.72 million who live alone have nobody nearby; no family is needed
- What a town lived through, in the voices of those who lived it

Who pays

Families
A subscription, against a ¥220,000 one-off book

Municipalities and care providers
Preventive care and reminiscence programmes

Cost to serve
≈ ¥570 a month for a daily conversation

Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.

Thank you

